COMPUTATIONAL ELECTRODYNAMICS

Jackson Electrostatics Survival Notes

This version is for studying, not showing off. It keeps the core electrostatics machinery visible: what equation I am solving, what region I am solving it in, what the boundary is doing, and which method is actually justified.

PREPARED FOR

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MAIN SOURCES

Jackson plus the notebooks, notes, formula sheets, and problems already in this repo

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The big idea

Electrostatics starts locally and ends globally. Locally I have Maxwell's equations. Globally I have a boundary-value problem. If I lose track of the physical region or the boundary conditions, the algebra may still look fine while the answer is wrong.

1. The actual structure of the subject

In electrostatics,

```
div E = rho / eps0
curl E = 0
E = -grad phi
```

So the potential satisfies Poisson's equation:

```
Laplace(phi) = -rho / eps0
```

In a source-free region this reduces to Laplace's equation:

```
Laplace(phi) = 0
```

The useful summary is:

- Poisson tells me how charge creates local curvature in the potential.
- Laplace governs charge-free regions.
- The PDE alone is never the whole problem.

- The physical solution is chosen by boundary conditions and asymptotic behavior.

2. What I should identify before touching algebra

1. Physical region: where do I actually need the solution?
2. Source structure: where is rho nonzero?
3. Boundary data: is phi specified, is the normal derivative specified, or is the conductor grounded?
4. Far-field behavior: what is the potential supposed to do at infinity?

Two common mistakes are solving Poisson in a region that is actually source-free, and using the right PDE with the wrong boundary condition.

3. Poisson's equation means local curvature

```
d^2 phi / dx^2 = -rho(x) / eps0
```

In 1D this gives a clean sign test:

- positive rho means downward curvature
- negative rho means upward curvature
- zero rho means the potential is linear in that region

The global warning is important: source-free does not mean zero potential. The region between conducting plates is charge-free and still supports a nontrivial field because the boundary forces it.

Main anchors here: the repo Poisson and Laplace notebooks plus the Poisson formula sheet.

4. Laplace's equation is rigid

Harmonic functions do not behave like arbitrary smooth functions. They are pinned down by the boundary. That is why source-free electrostatics often feels cleaner but more constrained.

Things worth remembering

- mean-value behavior
- no arbitrary interior extrema
- strong uniqueness properties
- superposition still works because the equation is linear

Radial source-free forms

```
spherical:  phi(r) = A + B/r
```

```
cylindrical: phi(s) = A ln(s) +
B
```

5. Boundary conditions are part of the physics

The PDE gives a family of solutions. The boundary picks the physical one.

- Dirichlet data: phi is known on the boundary
- Neumann data: the normal derivative is known on the boundary
- Grounded conductor: phi = 0 on the conductor
- Isolated conductor: equipotential, but not necessarily zero

If a surface charge density sigma is present, the normal field jumps:

```
E2_normal - E1_normal = sigma / eps0
```

When a solution looks almost right but fails on the boundary, it is wrong.

6. Uniqueness is what makes the toolkit legal

If I build a candidate potential that solves Poisson or Laplace in the physical region, satisfies the required boundary conditions, and has the right behavior at infinity, then uniqueness says that candidate is the physical answer.

That single fact is what makes the method of images rigorous and what makes Green-function constructions trustworthy.

7. Coulomb's law is already Green-function language

```
Laplace(G(r,r')) = -4 pi delta^3(r-r')
```

```
G(r,r') = 1 / |r-r'|
```

```
phi(r) = (1 / 4 pi eps0) integral rho(r') / |r-r'| d^3r'
```

The Coulomb kernel is the free-space fundamental solution of the Laplacian. Once boundaries appear, the operator stays the same but the correct Green function changes.

Main anchors here: the Green's functions notebook and formula sheet.

8. Why the delta function keeps showing up

```
rho(r) = q delta^3(r-r0)
```

```
Laplace(1/r) = -4 pi delta^3(r)
```

Away from the source point, 1/r is harmonic. At the source, it reproduces the correct singular charge in the distributional sense. That is exactly why the Coulomb potential solves Poisson's equation for a point charge.

9. The Green-function representation formula

```
phi(r) =
(1 / 4 pi eps0) integral_V rho(r') G(r,r') d^3r'
- (1 / 4 pi) integral_S [ G dphi/dn' - phi dG/dn' ] da'
```

The structure matters more than memorizing every sign on day one:

- one term carries the source information
- one term carries the boundary information
- the Green function is the object that knows the geometry

For Dirichlet problems it is often smart to pick a Green function that vanishes on the boundary.

10. The method of images is a boundary trick, not magic

1. Choose image charges in the excluded region.
2. Build a candidate potential.
3. Check the boundary exactly.
4. Check that only the real charges create singularities in the physical region.
5. Invoke uniqueness.

The image charges are not physical. They are a way to encode the boundary.

11. Example: point charge above a grounded plane

```
real charge:  q at (0,0,a)
image charge: -q at (0,0,-a)
```

```
phi(r) =
(1 / 4 pi eps0)
```

```
[ q / |r-r0| - q / |r-r0*| ]
```

Why it works:

- the physical region z > 0 contains only the real singularity
- on z = 0 the two distances match, so phi = 0 exactly
- the potential decays correctly at infinity

```
F = (1 / 4 pi eps0) q^2 / (2a)^2
```

Direction: toward the plane.

Main anchors here: the method-of-images notebook, formula sheet, and the grounded-sphere animation code.

12. Example: point charge outside a grounded sphere

Let a grounded conducting sphere of radius R sit at the origin and place a real charge q a distance a from the center, with a > R.

```
q' = -q R / a
b  = R^2 / a
```

The image charge sits on the same axis inside the sphere. The important pattern is:

- curved boundary means the image changes both position and strength
- the image still lies outside the physical region
- the sphere surface becomes an equipotential with phi = 0

This is a geometry-dependent Green-function construction in disguise.

13. Multipole expansion: how the far field forgets details

```
phi(r) ~ (1 / 4 pi eps0) [ Q/r + (p . rhat)/r^2 + ... ]
```

The far field remembers the coarse structure first:

- monopole term: total charge
- dipole term: first-order charge separation
- higher terms: finer angular structure

The more symmetry the source has, the more low-order moments vanish.

14. Dielectrics: same electrostatics, better bookkeeping

```
D = eps0 E + P
div D = rho_free
```

A dielectric does not kill the field like a conductor. It polarizes, which means bound charge appears and the accounting changes.

- free charge is what I put in explicitly
- bound charge comes from polarization
- boundary conditions must track the right quantity

15. A workflow that actually helps

1. Sketch the geometry.

2. Label the physical region.
3. Mark where charge really exists.
4. Decide whether each region uses Poisson or Laplace.
5. Write the boundary conditions separately.
6. Check symmetry honestly.
7. Choose the method only after that.

16. Fast checks that catch bad answers

- If the region is source-free, the candidate must be harmonic there.
- If there is a conductor, the surface must be equipotential.
- If the problem is grounded, the boundary value must be exactly zero.
- If it is an image solution, the images must stay outside the physical region.
- If the far field is wrong, the whole answer is suspect.
- If the units fail, stop and back up.

17. What I would want on one sheet before an exam

- E = -grad phi
- Laplace(phi) = -rho / eps0
- source-free does not mean zero potential
- uniqueness is the reason image methods and Green functions work
- 1 / |r-r'| is the free-space Green function
- Laplace(1/r) = -4 pi delta^3(r)
- grounded plane: mirror with charge -q
- grounded sphere: q' = -qR/a and b = R^2/a
- far field starts with monopole, then dipole
- in dielectrics, keep free and bound charge separate

References: Jackson 3rd ed.; repo notebooks on Poisson, Laplace, Green's functions, method of images, multipoles, and dielectrics; repo formula sheets and problem sets.